

AI-Driven Adaptive Anomaly Detection and Control for IoT Cyber-Physical Systems

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Abstract

This paper proposes ADAPT-IoT, an AI-driven adaptive control framework for secure and resilient Internet-of-Things cyber-physical systems. The framework integrates edge inference, anomaly detection, predictive analytics, security risk assessment, feedback control, cloud coordination, and human supervision. A discrete-time dynamical model is formulated with disturbance and anomaly terms. The contribution is theoretical and architectural, providing a mathematically grounded framework for real-time resilient control without claiming empirical results.

Keywords: Artificial Intelligence, Internet of Things, Cyber-Physical Systems, Dynamical Systems, Control Systems, Anomaly Detection, Machine Learning, Edge AI, Predictive Control, Cybersecurity

AMS Subject Classifications: 68T01, 68T07, 68M10, 93C40, 93C55, 93D05, 93D20, 93E10.

1. Introduction

The growth of Internet-of-Things (IoT) deployments has transformed many engineered systems into cyber-physical systems (CPS) in which sensing, computation, networking, and actuation are tightly connected. Industrial automation, water treatment, smart grids, connected transportation, autonomous platforms, and healthcare monitoring increasingly depend on distributed sensors, edge gateways, cloud services, embedded controllers, and human supervision. NIST distinguishes IoT from CPS while emphasizing their close relationship: IoT highlights connectivity and data exchange, whereas CPS highlights the closed-loop coupling between computation and physical processes [15], [16].

This coupling increases operational value but also increases risk. A corrupted sensor value, delayed communication packet, software fault, or adversarial input may influence a controller and propagate into unsafe physical behavior. Static monitoring and fixed-gain controllers remain useful in stable and well-understood settings, but they are insufficient for heterogeneous, nonstationary, and security-sensitive IoT CPS. The same plant may experience normal regime changes, sensor drift, cyberattacks, actuator degradation, and communication loss. A static threshold can either miss subtle anomalies or generate excessive false alarms. A controller tuned for nominal operation can continue applying unsafe control actions after measurements have been compromised.

Artificial intelligence offers a route to more adaptive operation. Machine learning and deep learning can model high-dimensional temporal patterns, identify multivariate deviations, reconstruct normal behavior, and estimate near-future risk [6], [21], [37]. Edge AI can move inference closer to the physical process and reduce dependence on cloud communication [4], [20], [31]. Control theory contributes rigorous notions of stability, constraints, adaptation, and fault tolerance [1], [2], [23], [29]. The challenge is not the

absence of individual methods; it is their integration into a coherent, trustworthy, and safe closed-loop architecture.

The research gap addressed here is the lack of a unified framework connecting anomaly detection, predictive analytics, security risk assessment, adaptive control, edge-cloud coordination, and human supervision. Many anomaly-detection systems produce alarms without changing controller behavior. Many control systems optimize performance without explicitly accounting for cyber risk. Many edge-computing architectures reduce latency without providing safety guarantees. This paper proposes ADAPT-IoT to interrelate these ideas through a dynamical and control-systems lens.

The contributions of this paper are: (i) a novel framework named ADAPT-IoT for secure and resilient IoT CPS; (ii) a mathematical model incorporating disturbances, anomaly signals, risk scores, and constrained control objectives; (iii) an AI model design combining shallow learning, recurrent networks, transformers, autoencoders, graph neural networks, and safe reinforcement-learning-assisted adaptation; (iv) pseudocode for real-time anomaly detection, adaptive control, edge inference, fault recovery, and security-aware response; and (v) a theoretical analysis of safety, explainability, limitations, and ethical deployment.

Research Questions

- RQ1. How can AI improve anomaly detection in IoT cyber-physical systems?
- RQ2. How can adaptive control improve system resilience under faults and attacks?
- RQ3. How can edge AI reduce latency in real-time control environments?
- RQ4. How can AI-based control systems remain explainable and safe?
- RQ5. What are the main limitations and future research directions?

2. Background and Related Work

IoT, CPS, and dynamical systems

Cyber-physical systems are best understood as dynamical systems whose physical evolution depends on computation, communication, sensing, and actuation. This view is essential because cyber events can have physical consequences. CPS security surveys emphasize that attacks on sensors, controllers, and communication channels must be analyzed together with physical dynamics, process constraints, timing assumptions, and safety requirements [13], [18], [25].

A generic discrete-time process is represented by a state, a control input, an observation, and disturbances. Classical dynamical-systems analysis provides the vocabulary for equilibrium, stability, observability, controllability, and robustness. In IoT CPS, these properties are complicated by heterogeneous sampling rates, uncertain network latency, device constraints, and adversarial manipulation.

Control, adaptive control, and predictive control

Control theory provides established mechanisms for stabilizing and optimizing physical systems. PID control remains common in engineering because it is simple and interpretable, yet it is weak when constraints, nonlinearity, and changing operating regimes must be handled explicitly. Model predictive control (MPC) addresses these limitations by solving a constrained finite-horizon optimization problem at each sampling time [23], [24], [29].

Adaptive control modifies parameters online when plant dynamics change [2]. Fault-tolerant control adds diagnosis, isolation, and controller reconfiguration so that acceptable performance can continue after faults [1]. These ideas are directly relevant to secure IoT CPS because cyberattacks may appear to the controller as faults, disturbances, or corrupted observations. However, control logic must distinguish benign regime change from malicious manipulation whenever possible.

AI-based anomaly detection

Anomaly detection identifies observations or trajectories that deviate from an expected pattern [6]. In IoT CPS, anomalies may correspond to process faults, sensor degradation, false data injection, replay attacks, denial of service, actuator faults, or unmodeled environmental changes. Classical machine-learning models such as support vector machines and random forests can be useful when informative features are available. Deep models are attractive when temporal dependence and nonlinear coupling dominate.

Recurrent models such as LSTM and GRU capture sequential dependencies [7], [17]. Autoencoders and stochastic recurrent models support unsupervised detection by learning normal reconstruction or latent dynamics [3], [33]. Transformers capture longer-range temporal context and have been adapted to time-series forecasting and anomaly detection [35], [36], [38]. Graph neural networks exploit relationships among sensors and actuators and are particularly useful when physical topology or learned dependency structure matters [8].

Edge AI, cybersecurity, and trustworthy operation

Cloud-centric AI can be effective for heavy training and fleet-level analytics, but real-time control often requires decisions close to the plant. Edge AI reduces communication delay, bandwidth demand, and dependence on wide-area network availability [4], [20], [31]. The cost is constrained computation, limited memory, energy consumption, update complexity, and a larger edge attack surface.

Cybersecurity threats for IoT CPS include false data injection, sensor spoofing, denial-of-service attacks, adversarial examples, model poisoning, and communication replay [5], [10], [11], [14], [26], [28], [32]. Trustworthy AI guidance further emphasizes reliability, safety, security, resilience, transparency, accountability, and human oversight [27]. ADAPT-IoT is designed around these requirements rather than treating AI accuracy as the sole objective.

Comparative Synthesis

Table 1: Comparative Synthesis of Traditional Static Design and ADAPT-IoT

Dimension	Traditional static design	ADAPT-IoT design
Monitoring	Fixed thresholds and rule-based alarms	Learned multivariate anomaly scoring with uncertainty
Control	Nominal PID or fixed controller settings	Risk-aware predictive and adaptive control
Computation	Cloud-centric analytics or isolated controllers	Hybrid edge-cloud inference and coordination
Security	Perimeter defense and signatures	Behavioral, physics-aware, and risk-aware response
Supervision	Operator logs and manual intervention	Human dashboard with confidence, explanation, and override

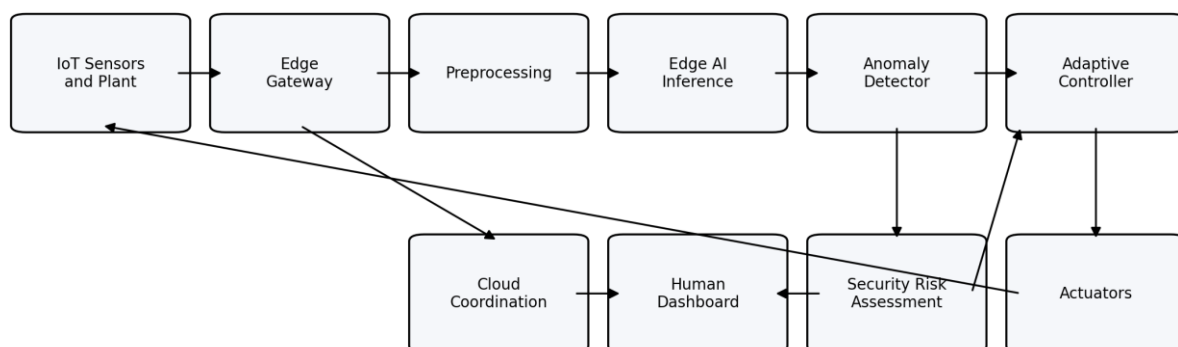
3. Problem Definition

Modern IoT cyber-physical systems generate continuous, dynamic, heterogeneous, and security-sensitive data. Traditional control and monitoring approaches often fail to detect anomalies early, adapt to changing environments, and respond safely in real time. The core problem is to transform uncertain and possibly adversarial sensor streams into control actions that preserve safety and stability while maintaining acceptable performance.

The central research problem can therefore be stated as follows: how can an IoT CPS detect anomalous behavior, estimate cyber-physical risk, and adapt its control policy in real time while preserving safety, stability, explainability, and operational resilience?

ADAPT-IoT is an AI-driven adaptive control framework for secure and resilient IoT cyber-physical systems. It treats sensing, anomaly detection, cyber-risk assessment, predictive analytics, and control as components of one supervisory loop. The framework is modular so that domain-specific plants can instantiate different AI models, controllers, and security policies while preserving the same architectural principles.

Figure 1: ADAPT-IoT Architecture

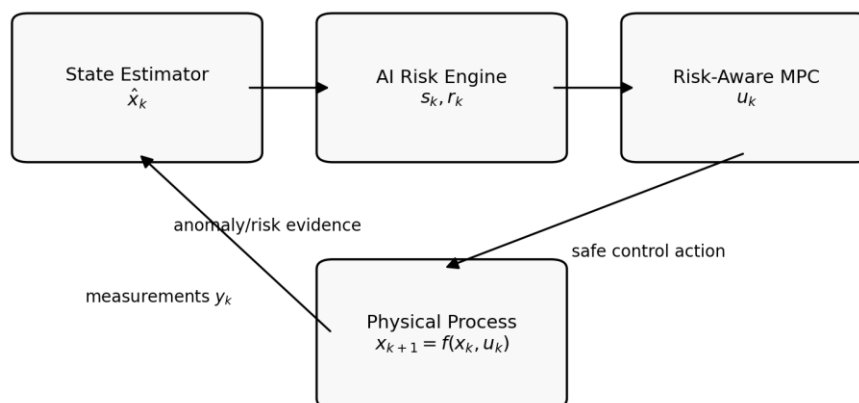
**Table 2: ADAPT-IoT Layer Description**

Layer	Function	Typical output
IoT sensing	Collects plant, actuator, controller, and network telemetry.	Raw multivariate streams
Preprocessing	Aligns time, handles missing values, filters noise, and constructs windows.	Feature tensors

Layer	Function	Typical output
Edge AI inference	Runs lightweight detection and forecasting close to the plant.	Local predictions and latency logs
Anomaly detection	Computes ensemble anomaly scores and root-cause candidates.	Score, label, confidence
Predictive analytics	Forecasts near-future state, risk, and degradation trends.	Predicted trajectories
Adaptive control	Computes control under safety and risk constraints.	Control action
Security assessment	Maps anomaly evidence to operational cyber risk.	Risk score and threat class
Feedback loop	Applies actions and updates the state estimate.	Closed-loop correction
Cloud coordination	Supports heavy training, model registry, and fleet analytics.	Updated models and policies
Human dashboard	Provides explanations, alarms, audit logs, and override.	Supervisory decision support

The key design principle is that an anomaly score is not merely an alarm. In ADAPT-IoT, anomaly evidence changes the control regime, modifies trust in measurements, tightens constraints, and can trigger safe-hold operation. This gives the architecture a direct connection between detection and physical resilience.

Figure 2: Risk-Aware Feedback Control Loop



5. Mathematical Formulation

Let the IoT CPS be modeled in discrete time. The state is x_k in $\mathbb{R}^n$, the control input is u_k in $\mathbb{R}^m$, and the measured output is y_k in $\mathbb{R}^p$. The system evolves according to a nonlinear mode-dependent model with disturbances and anomaly signals:

$$x_{k+1} = f_{\sigma_k}(x_k, u_k, \theta_k) + E_w w_k + E_a a_k,$$

$$y_k = h_{\sigma_k}(x_k) + v_k + F_a a_k.$$

Here w_k is process disturbance, v_k is measurement noise, a_k is an anomaly or attack signal, θ_k represents slowly varying parameters, and σ_k is the operating mode. The anomaly term is allowed to enter both the dynamics and the observation equation because attacks may target actuators, sensors, or communication links [11], [28].

The detector receives a sliding-window feature tensor:

$$zeta_k = \Phi(y_{\{k-L:k\}}, u_{\{k-L:k\}}, \Delta y_{\{k-L:k\}}, G_k),$$

where L is the window length and G_k is a sensor-actuator dependency graph. The anomaly score is computed as:

$$s_k = D_{\Theta}(zeta_k), 0 \leq s_k \leq 1.$$

To avoid a brittle global threshold, ADAPT-IoT uses a context-aware threshold:

$$\tau_k = \tau_0 + \lambda_u U_k + \lambda_c C_k,$$

where U_k is model uncertainty and C_k captures operating context. A risk score combines anomaly evidence, attack likelihood, and sensor trust:

$$r_k = \alpha s_k + \beta \Pr(\text{attack} | zeta_k) + \gamma \text{TrustPenalty}_k.$$

The adaptive controller solves a constrained receding-horizon problem:

$$\min_{\{u_{\{k:k+H-1\}}\}} J_k$$

subject to predicted dynamics and safety constraints:

$$x_{\{k+i+1|k\}} = \hat{f}(x_{\{k+i|k\}}, u_{\{k+i|k\}}),$$

$$x_{\{k+i|k\}} \in X_{\text{safe}}(r_k), \quad u_{\{k+i|k\}} \in U_{\text{safe}}(r_k).$$

A representative objective is:

$$J_k = \sum_{i=0}^{H-1} (\|x_{\{k+i|k\}} - x_{\text{ref}}\|_Q^2 + \|u_{\{k+i|k\}}\|_R^2 + \|\Delta u_{\{k+i|k\}}\|_S^2 + \lambda_r r_{\{k+i|k\}}) + V_f(x_{\{k+H|k\}}).$$

The risk term converts detection into control-relevant information. Higher risk tightens safety constraints, reduces trust in suspicious measurements, and can force a safe mode. The supervisory mode is:

$$\sigma_k = \text{nominal}, \quad \text{if } s_k < \tau_1 \text{ and } r_k < \rho_1,$$

$$\sigma_k = \text{resilient}, \quad \text{if } \tau_1 \leq s_k < \tau_2 \text{ or } \rho_1 \leq r_k < \rho_2,$$

$$\sigma_k = \text{safe-hold}, \quad \text{if } s_k \geq \tau_2 \text{ or } r_k \geq \rho_2.$$

A sufficient practical stability condition can be stated using a Lyapunov function V :

$$V(x_{\{k+1\}}) - V(x_k) \leq -\alpha \|x_k\|^2 + \beta \|w_k\|^2 + \gamma \|a_k\|^2.$$

This condition expresses that the closed loop dissipates state energy up to bounded effects of disturbances and attacks. In attack-free operation with recursive feasibility and bounded disturbances, the controller is expected to preserve practical stability. Under high risk, the objective shifts from optimal performance to safe containment.

6. AI Model Design

ADAPT-IoT is model-agnostic but uses explicit model roles. No single model is assumed to dominate across all plants, attacks, and hardware conditions.

Table 3: AI Model Families for ADAPT-IoT

Model family	Role in ADAPT-IoT	Strength	Limitation
Random Forest	Low-cost edge or fallback detector	Fast and interpretable on engineered features	Limited temporal memory

Model family	Role in ADAPT-IoT	Strength	Limitation
Support Vector Machine	Classical anomaly classifier	Useful margins in moderate data regimes	Scaling and calibration challenges
LSTM / GRU	Temporal anomaly detection and forecasting	Captures sequence dependence	Training cost and drift sensitivity
Transformer	Long-context forecasting and anomaly reasoning	Models global temporal structure	Expensive for constrained edge devices
Autoencoder	Unsupervised reconstruction-based detection	Useful when labels are scarce	Threshold calibration can be unstable
Graph Neural Network	Sensor-relation and topology-aware detection	Detects dependency violations and aids root cause analysis	Requires process graph or graph learning
Reinforcement Learning	Policy adaptation in simulation or shadow mode	Learns sequential control adjustments	Unsafe without constraints and supervisory shielding

The ensemble score combines shallow, reconstruction, temporal, and graph-based evidence:

$$s_k = \omega_1 s_k^{\text{shallow}} + \omega_2 s_k^{\text{recon}} + \omega_3 s_k^{\text{temporal}} + \omega_4 s_k^{\text{graph}}, \quad \sum \omega_i = 1, \omega_i \geq 0.$$

The ensemble provides two benefits. First, it reduces dependence on a single failure mode. Second, disagreement among detectors provides an uncertainty signal that can be fed into the adaptive threshold and controller. Explanations can be produced through residual ranking, feature attribution, and graph attention, but these explanations should be treated as decision support rather than mathematical proof of causality [21].

7. Algorithms

Real-time anomaly detection

Algorithm 1: RealTimeAnomalyDetection

Input: sensor stream y_k , control stream u_k , detector set M , window length L

Output: anomaly label, anomaly score s_k , uncertainty U_k , explanation

1. Update rolling window W_k with y_k and u_k .
2. Align timestamps, filter noise, normalize values, and extract features.
3. For each detector m in M , compute detector-specific score $s_{\{m,k\}}$.
4. Aggregate scores into ensemble score s_k .
5. Estimate uncertainty U_k from disagreement and calibration.
6. Compute adaptive threshold τ_k .
7. If $s_k \geq \tau_k$, label the event anomalous; otherwise label it nominal.
8. Generate root-cause candidates from residuals, feature attribution, and graph relations.
9. Send score, uncertainty, and explanation to the risk-aware controller.

Adaptive control decision-making

Algorithm 2: AdaptiveControlDecision

Input: state estimate $\hat{x}_k$, anomaly score s_k , risk score r_k

Output: control action u_k

1. If risk is low, solve nominal MPC.
2. If risk is moderate, tighten state and input constraints and solve resilient MPC.
3. Reduce estimator trust in suspicious sensors.
4. If risk is high, switch to safe-hold or degraded operating mode.
5. Alert the human supervisor and preserve an audit record.
6. Apply only the first feasible control action and repeat at the next sampling time.

Edge AI inference and recovery

Algorithm 3: EdgeInferenceAndRecovery

Input: feature window W_k , edge budget, communication status, model registry

Output: inference result and recovery action

1. Estimate latency, memory, and energy budget.
2. If the lightweight edge model satisfies the deadline, run local inference.
3. Else if network risk is low, offload selected features or embeddings to the cloud.
4. Else run a signed fallback detector locally.
5. If persistent abnormal residuals occur, isolate suspicious channels.
6. Re-estimate state with low-trust sensors masked or down-weighted.
7. Reconfigure controller; if safety cannot be guaranteed, enter safe-hold mode.

Security-aware response and safe reinforcement learning

Algorithm 4: SecurityAwareResponse

Input: attack class probability, proposed control action, safety constraints

Output: approved control action

1. Authenticate data source, model version, and control message.
2. If false data injection is suspected, increase estimator redundancy and tighten constraints.
3. If denial of service is suspected, switch to local autonomous operation.
4. If model poisoning is suspected, freeze updates and use a signed fallback model.
5. Approve the control action only if safety constraints are satisfied.

Algorithm 5: SafeRLPolicyAdaptation

Input: digital twin, safety envelope, reward function

Output: candidate adaptive policy

1. Train reinforcement-learning policies only in simulation or a digital twin.
2. Randomize disturbances, faults, delays, and attack scenarios.
3. Penalize constraint violations and unsafe actions.
4. Validate candidate policies against safety envelopes.
5. Deploy only in shadow mode or behind an MPC safety shield.

8. Security and Safety Analysis

ADAPT-IoT treats cybersecurity events as control-relevant disturbances. A cyberattack matters not merely because it compromises information, but because compromised information can induce unsafe physical decisions. Security analysis must therefore connect attack class, system state, estimator trust, and controller mode.

Table 4: Security Threats and ADAPT-IoT Mitigations

Threat	Primary target	Control consequence	ADAPT-IoT mitigation
False data injection	Sensors and estimators	Biased state estimate and unsafe control	Residual checks, graph consistency, trust-weighted estimation
Sensor spoofing	Sensing layer	False local perception	Cross-sensor validation and redundancy
Denial of service	Communication layer	Delayed or missing feedback	Local edge fallback and degraded control
Replay attack	Data channel	Stale measurements or commands	Freshness checks and authenticated messages
Adversarial examples	AI detector inputs	Misclassification of unsafe events	Ensemble disagreement and conservative thresholds
Model poisoning	Training or update pipeline	Corrupted detector or policy	Signed updates, quarantine, shadow validation
Actuator compromise	Control channel	Unsafe physical action	Safety filter, constraints, and safe-hold mode

Safety is handled through layered defense. The detector provides evidence, the security module converts evidence into risk, the controller tightens constraints or changes modes, and the human dashboard provides supervision. Safe reinforcement learning is not given direct unconstrained control authority. Instead, learned policies are trained in simulation and shielded by model predictive or supervisory safety logic [12], [30], [34].

9. Findings and Theoretical Implications

Table 5: Findings and Theoretical Implications

Finding	Technical explanation	Practical implication
AI improves detection when temporal and relational context are modeled.	CPS anomalies often appear as coordinated deviations across multiple channels rather than isolated threshold violations.	Use ensembles that combine time-series, reconstruction, and graph evidence.
Adaptive control improves resilience only if anomaly evidence changes control behavior.	An alarm that does not affect estimation, constraints, or mode switching cannot by itself prevent unsafe actuation.	Connect anomaly scores to risk-aware MPC and safe-hold policies.
Edge AI reduces latency but cannot fully replace cloud coordination.	Edge inference shortens the decision path; cloud resources remain valuable for training, fleet analytics, and updates.	Use hybrid edge-cloud deployment with signed model updates.

Finding	Technical explanation	Practical implication
Explainability is part of safety.	Operators need confidence, uncertainty, and root-cause information when AI affects physical control.	Include dashboards, audit logs, and override mechanisms.
Integration is the main contribution.	Strong components exist separately in anomaly detection, edge AI, and control, but closed-loop integration remains challenging.	Evaluate future systems as cyber-physical feedback loops, not isolated classifiers.

10. Discussion

ADAPT-IoT improves dynamic control systems by turning high-dimensional sensor histories into decision-relevant state and risk estimates. This is important because IoT CPS anomalies often unfold over time and across correlated channels. A single threshold may miss early signals; a single deep model may be difficult to explain or deploy at the edge. The framework therefore separates detection, uncertainty, risk, and control while allowing the modules to exchange information.

The central trade-off is between accuracy, latency, safety, and energy consumption. Large sequence models can improve anomaly coverage but increase inference cost. Edge inference lowers latency but constrains model size. Conservative thresholds reduce false negatives but may increase false alarms. Reinforcement learning can optimize adaptation but creates safety concerns if deployed without shielding. ADAPT-IoT addresses these tensions through hybrid placement, ensemble uncertainty, risk-aware MPC, and safe fallback modes.

The framework is suitable for industrial IoT, smart cities, smart grids, healthcare devices, autonomous platforms, and water or energy infrastructure. Each domain requires different latency budgets, safety envelopes, privacy rules, and operator workflows. Therefore, ADAPT-IoT should be viewed as a reference architecture rather than a single universal implementation.

11. Limitations

- The paper is theoretical and architectural; it does not report new empirical results.
- Formal verification of the full AI-control supervisor remains future work.
- AI models may not generalize across plants, devices, or operating regimes.
- Edge hardware may limit feasible model size, memory use, and update frequency.
- Security evaluation is incomplete without domain-specific threat modeling and red-team assessment.
- Human-in-the-loop usability and trust calibration require separate validation.
- No claim is made that ADAPT-IoT is optimal for every IoT CPS domain.

12. Ethical and Societal Impact

AI-driven IoT CPS control raises significant ethical issues. Continuous sensing can create privacy and surveillance risks. Automated control in safety-critical environments can cause harm if confidence is miscalibrated, if rare hazards are absent from training data, or if operators over-trust opaque outputs. Edge AI can reduce raw-data transmission, but it does not remove the need for data minimization, access control, auditability, and accountability.

Responsible deployment requires privacy-preserving data handling, security-by-design, documented safety envelopes, incident logging, human override, and post-deployment monitoring. AI should support operators rather than silently replacing safety responsibility. In high-risk sectors, ADAPT-IoT should be reviewed under applicable engineering, cybersecurity, privacy, and safety standards before deployment.

13. Conclusion

This paper proposed ADAPT-IoT, an AI-driven adaptive control framework for secure and resilient IoT cyber-physical systems. The framework integrates sensing, preprocessing, edge inference, anomaly detection, predictive analytics, security risk assessment, adaptive control, feedback, cloud coordination, and human supervision. A discrete-time dynamical model was formulated with disturbance and anomaly terms, and a risk-aware predictive-control objective was defined under safety constraints.

The central contribution is the integration of AI-based anomaly detection with adaptive and predictive control. ADAPT-IoT does not treat anomalies as detached alerts; it uses anomaly and risk evidence to alter estimator trust, tighten constraints, switch operating modes, and preserve safe operation. The framework provides a rigorous foundation for future implementation, formal verification, and domain-specific validation of trustworthy AI control in next-generation IoT CPS.

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